

BACHELORS WITH STATISTICS AS MAJOR
6th SEMESTER

STS622N STATISTICS _ STATISTICAL COMPUTING-I

CREDITS: 3 THEORY +1 PRACTICAL
THEORY: 45 HOURS & PRACTICAL: 30 HOURS

COURSE OBJECTIVES:

To express the students to the real-life skill for statistical computing analysis and graphical interpretation using software skill. Hands on training on the real-life problems can be done on any one of the statistical software/excel to enhance data analysis skills.

COURSE OUTCOMES:

After completing this course, students must be able:

- *To understand the Graphical representation of data sets.*
- *To understand the descriptive statistics and Correlation analysis.*
- *To understand the usages of statistical and computational software's.*

THEORY: 03 CREDITS

UNIT-I

Introduction to Computers: Historical evolution of Computers. Generations of Computers. Classification of Computers, Applications of Computers, Computer Memory: Primary and Secondary Memory, Hardware: CPU, I/O Devices.

UNIT II

Word Processing: Creating and Saving a Document, Editing the Text: Printing, Saving and Importing Documents. Basics of Excel: Data Entry, Data sorting, Data validation, filtering of data, built in statistical and Mathematical Functions in Excel.

UNIT III

Diagrammatic and Graphical representation of Data in excel: bar diagrams and pie diagrams, Histogram, frequency Polygon, frequency curve, Working with data manipulation, descriptive statistics, simple Correlation.

PRACTICAL: (01 CREDIT) USING STATISTICAL / COMPUTATIONAL SOFTWARE'S

- I. Problems based on graphical representation of data by bar diagrams and pie diagrams
- II. Problems based on graphical representation of data by Histogram
- III. Problems based on graphical representation of data by Frequency polygons.
- IV. Problems based on calculation of Measures of Central Tendency.
- V. Problems based on calculation of Measures of Dispersion.
- VI. Problems based on calculation of Correlation.

BOOKS RECOMMENDED:

1. Moore, D.S. and McCabe, G.P. and Craig, B.A.: Introduction to the Practice of Statistics, W.H. Freeman, (2014).
2. Cunningham, B.J: Using SPSS: An Interactive Hands-on approach, (2012).
3. Cho, M, J., Martinez, W.L.: Statistics in MATLAB: A Primer, Chapman and Hall/CRC, (2014).
4. E.J. Dudewicz and S.N. Mishra: Modern Mathematical Statistics, Willy, Int'l Students edition, (1988).
5. John Verzani.: Using R for Introductory Statistics. Chapman & Hall/CRC, (2005).
6. S.C. Gupta and V.K. Kapoor: Fundamentals of Mathematical Statistics, Sultan Chand & Sons, (2012).